Sungjun Heo

Ulsan, Republic of Korea https://rml.unist.ac.kr/ Sungjun Heo

Education

Integrated M.S.–Ph.D. **Ulsan National Institute of Science and Technology (UNIST)** Mar 2024 -

- Graduate School of Artificial Intelligence
- Robotics & Mobility Laboratory of Prof.J Jeon
- Research Field : End-to-End Autonomous Driving, Reinforcement Learning based Planning, Robotics.
- GPA : 4.0 / 4.3

B.S **Ulsan National Institute of Science and Technology (UNIST)** Mar 2021 - Fed 2024

- Department of Electrical Engineering
- Early Graduation(by one year)
- GPA : 3.6 / 4.3 , Total credits : 127

Research Interest

Autonomous Driving(AD)

- RGB camera based End-to-End (E2E) autonomous driving
- Reinforcement learning based driving planning
- Low sensor fusion based planning (Spatiotemporal fusion model)

Reinforcement Learning(RL)

- Structure designing Hierarchical RL / Curriculum RL in online reinforcement learning
- Data optimization in offline reinforcement learning

Robot manipulation

- Long-horizon manipulation in collision-sensitive task
- Sim-to-Real / Real-to-Real action learning

Publications

Sungjun Heo and J. Jeon, **"Predicted State-Based Hierarchical Reinforcement Learning for Long-Term Decision Making in Urban Dynamic Scenarios"** - Proceeding in IEEE Int. Veh. Sym. (Flagship Conference of IEEE ITSS - IV 2025), pp. 1164–1170. Fed 2025

Sungjun Heo, H. Jeong, E. Kim, H. Lee, J. Lee, S. Lee, at el., **"Capacity-Controlled Enforcement of Sparse Traffic-Light Cues for Rule-Conformant End-to-End Driving"** - Under reviewing in IEEE International Conference on Systems, Man, and Cybernetics (SMC), (Flagship conference of the IEEE SMC Society - 2026 SMC) May 2025

Sungjun Heo and J. Jeon, **"One paper in Double blind review process"** - Under review in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), (Major Conference of IEEE RAS - IROS 2026) Sep 2025

H. Jeong, E. Kim, **Sungjun Heo**, S. Kim, S. Lee, J. Shin at el., **"Deployment-Oriented End-to-End Autonomous Driving: Enhancing Closed-Loop Stability with a Lightweight Framework"** - Accepted in IEEE Int. Veh. Sym. (Flagship Conference of IEEE ITSS - IV 2026) Oct 2025

Ongoing Publications, Project

Sungjun Heo, S. Yang and J. Jeon, **"BridgeFlow: Zero-Full-Demo Long-Horizon Manipulation via Transition-Conditioned Stage Composition"** Nov 2025 - Current

Awards and Honors

Academic Achievement Award 2021

- Undergraduate academic achievement excellence award in 2021

Undergraduate Research Award (Top 5 / All graduation students) Dec 2023

- Research Title : Autonomous Driving Waypoint-generation based on Vehicle Kinematics.

3rd place in Preliminary Round (3 / 16 teams) Mar 2025

- Preliminary Round of 2025 Hyundai Motor Group's Autonomous Driving Challenge
- RGB camera based End-to-End Mission-based Autonomous driving challenge in K-City

1st place in 2025 Hyundai Motor Group's Autonomous Driving Challenge (1 / 16 teams) Sep 2025

- Final 2025 Hyundai Motor Group's Autonomous Driving Challenge
- RGB camera based End-to-End Full-Self-Driving in K-City

Grand Prize - Brain To Society, U-Challenge Festival (1 / 96 teams) Dec 2025

- Brain To Society : Future Mobility part
- Title : Sensor-driven Hierarchical Information Fusion Transformer for BEV-based Maneuvering in Dense Multi-lane Environments

Work Experience

Paper Reviewer

- 3 Papers in 2025 IEEE INTELLIGENT VEHICLES SYMPOSIUM
- 2 Papers in 2025 IEEE International Conference on Intelligent Transportation Systems
- 2 Papers in 2026 IEEE INTELLIGENT VEHICLES SYMPOSIUM
- 1 Paper in 2026 IEEE/RSJ International Conference on INTELLIGENT ROBOTS & SYSTEMS

AI Innovation Park Internship Jul - Sep 2024

- Development of a time-series-based AI model for power prediction

Project

• Major Projects & Challenge

Preliminary Round : 2025 Hyundai Motor Group's Autonomous Driving Challenge Sep 2024 - Mar 2025

- Position : Team Member(model framework, dataset)
- RGB camera based End-to-End Mission-based Autonomous driving challenge in K-City

Final Round : 2025 Hyundai Motor Group's Autonomous Driving Challenge Apr 2025 - Oct 2025

- Position : Team Member(rule-based expert driving dataset, model framework)
- RGB camera based End-to-End Full-Self-Driving in K-City

Brain to Society : Future mobility Jul - Dec 2025

- Position : Team Leader
- Project Title : Sensor-driven Hierarchical Information Fusion Transformer for BEV-based Maneuvering in Dense Multi-lane Environments

• Academic Projects

Low-cost Robot Manipulation (Robotics: Science and Systems) Dec 2024

- Position : Team Member

- Paper Title :Demonstrating a Vision-Based AI Robot for Strategic Board Games
- Project page: link

3D Vision and Machine Cognition Jun 2025

- Position : Team Member
- Project Title : Traffic-Light and Boundary-Aware BEV Learning from Multi-Camera Images

2D Vision: Human Pose Estimation Jun 2025

- Title : SoftEdge-PoseContrast View Conditioned Human-Pose Classifier with Contrastive Learning